DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

Public reporting burden for this collection of information is estimated to average 110 hours per response, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information. Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden, to Washington Headquarters Services, Directorate for Information Operations and Reports, 1215 Jefferson Davis Highway, Suite 1204, Arlington, VA 22202-4302, and to the Office of Management and Budget, Paperwork Reduction Project (0704-0188), Washington, DC 20503.

1. TITLE	2.. IDENTIFICATION NUMBER
Motor Log Book	DI_MISC -81240

3. DESCRIPTION/PURPOSE

3.1 The Motor Log Book will be used to provide the acquiring field and depot suppport agencies with the condensed history and delivered status of each rocket motor.

3.2 The Motor Log Book will provide:

a. Deviations and waivers from model specifications during production requiring contractor or Defense Plant Representative Office (DPRO) acceptance.

(Continued on page 2)

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
911010	F/OO-ALC-LMAPM		

7. APPLICATION INTERRELATIONSHIP

7.1 This Data Item Description contains the format and content preparation instructions for the Motor Log Book generated by specific and discrete task requirements as delineated in the contract.

7.2 This DID replaces cancelled DI-L-3343.

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		F6681

10. PREPARATION INSTRUCTIONS

10.1 Format: The Motor Log Book shall have a title page and eight individual sections in accordance with 10.2 below. Formats may be in accordance with samples shown in Figures 1 through 8. The Log Book shall be standard letter size on bond paper or reproduced (8 1/2" x 11"). It shall be in a loose leaf binder with sections of the Motor Log Book separated by divider pages with title tabs.

10.2 Content:

a. Title page (first page) shall contain the following information:

(1) The words "Motor Log Book".
(2) Model Number.
(3) Part Number.
(4) Serial Number.
(5) Contract Number.
(6) National Stock Number.

b. Section One - Acceptance Record (DD Form 250).

c. Section Two - Quality Certification. This section shall contain the statement,

11. DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

ATCH 1

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Block 3, Description/Purpose (Continued)

b. Critical dimensions (X, Y and Z) for alignment of nozzle control units which are essential for depot repair activities.

c. Predicted motor performance such as impulse and thrust measurements required by Flight Analysis Personnel.

d. A list of Engineering Change Proposals (ECPs) accomplished during production and date of manufacture.

e. Propellant load date.

f. Motor weight.

g. Weight and balance and center of gravity information.

h. Propellent lot/batch manufacture information.

Block 10, Preparation instructions (Continued)

"This is to certify that contractual requirements for quality control and assurance have been complied with in the manufacture of these items. It has been determined that these items are in conformance with applicable specifications and any departures therefrom have been accepted by approved material review procedures or are listed as Class I departures for contractual action. Pertinent inspection and test reports are on file at this facility and available for examination. All departures/changes from the listed baseline are delineated in the appropriate section of the Motor Log. The environmental history of the motor has been examined and determined to be in conformance with the requirements of model specification. Non-destructive inspection has been performed to assure conformance to the requirements of the applicable motor specification.". The Certificate shall also contain the model number, part number, serial number, nomenclature, model specification, baseline per, date of manufacture (i.e., date of core removal), manufacturer's contract number, ECPs requiring field modification, accepted by, date, motor inspection number and lot number.

d. Section Three - Assembly Component and Serialization Record. Format may be in accordance with sample shown in Figure 1.

e. Section Four - Design Changes (from Baseline). Shall contain the model number, part number, serial number of the motor, any Class I design changes and the ECP delivery list.

f. Section Five - Waivers or Deviations. Shall contain the model number, part number, serial number of the motor, any departures acceptable outside of specification limits and supporting Nonconformance Report (NCR) documents.

g. Section Six - Weight and Center of Gravity. Formats may be in accordance with samples shown in Figures 2, 3, 4 and 5, based on motor configuration.

h. Section Seven - Predicted Motor Performance. Shall contain the motor serial number, parameters, values, model specification, prepared by and an approval signature by the contractor's quality control department.

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Block 10, Preparation Instructions (continued)

i. Section Eight - Inspection and Test Data Summary. Shall contain the results of tests and inspections. For example:

(1) Nozzle Data. Shall contain the serial number, part number, vendor, No Tare X, Y and Z readings, Tare X, Y and Z readings, fixed plane alignment and wedge washer locations as applicable.

(2) Pressure Transducer Calibration. Format may be in accordance with sample shown in Figure 6.

(3) Standardized Igniter Safe Arm Mechanism Inspection. Format may be in accordance with sample shown in Figure 7.

(4) Standardized Thrust Termination Switch Inspection Report. Format may be in accordance with sample shown in Figure 8.

(5) Transducer Electrical Test. Transducer electrical test results when tested in accordance with appropriate motor technical order or transducer specification. Test sheet shall contain transducer part number, serial number and part number and serial number of the test set. Contractor format acceptable upon government approval.

(6) Roll Control Valve Test. Roll control valve electrical test results when tested in accordance with appropriate motor technical order. Test sheet shall contain the serial number of the roll control valve and the part number and serial number of the test set. Contractor format acceptable upon government approval.

(7) Propellent Data Sheet. Shall contain motor serial number, contract number, type of propellent, batch and mix designation, date of cast and core removal date. A statement that all propellents used in the motor have been analyzed and conform to the appropriate specification shall be included.

(8) Radiographic Inspection Report. Shall contain results of radiographic inspection of the rocket motor. Contractor format acceptable upon government approval.

j. The use of the abbreviation "N/A" shall be a required entry for all areas that are "Non-Applicable".

k. Security classification and distribution markings shall conform to the requirements of the contract.

l. The Air Force serial number of the motor shall be displayed on each page of the Motor Log Book in addition to any contractor serial number used.

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Block 10. Preparation Instructions (Continued)

MOTOR SER NO _____

AF CONTRACT _____

DATE _____

VENDOR/ SUPPLIER	COMPONENT/ PART NAME	PART/DRAWING NUMBER	DASH NUMBER	REVISION NUMBER	SERIAL NUMBER	DATE OF MANUFACTURE (TIME CHANGE ITEMS ONLY)

CERTIFIED BY: _____ QUALITY CONTROL

FIGURE 1. Sample assembly component and serialization record

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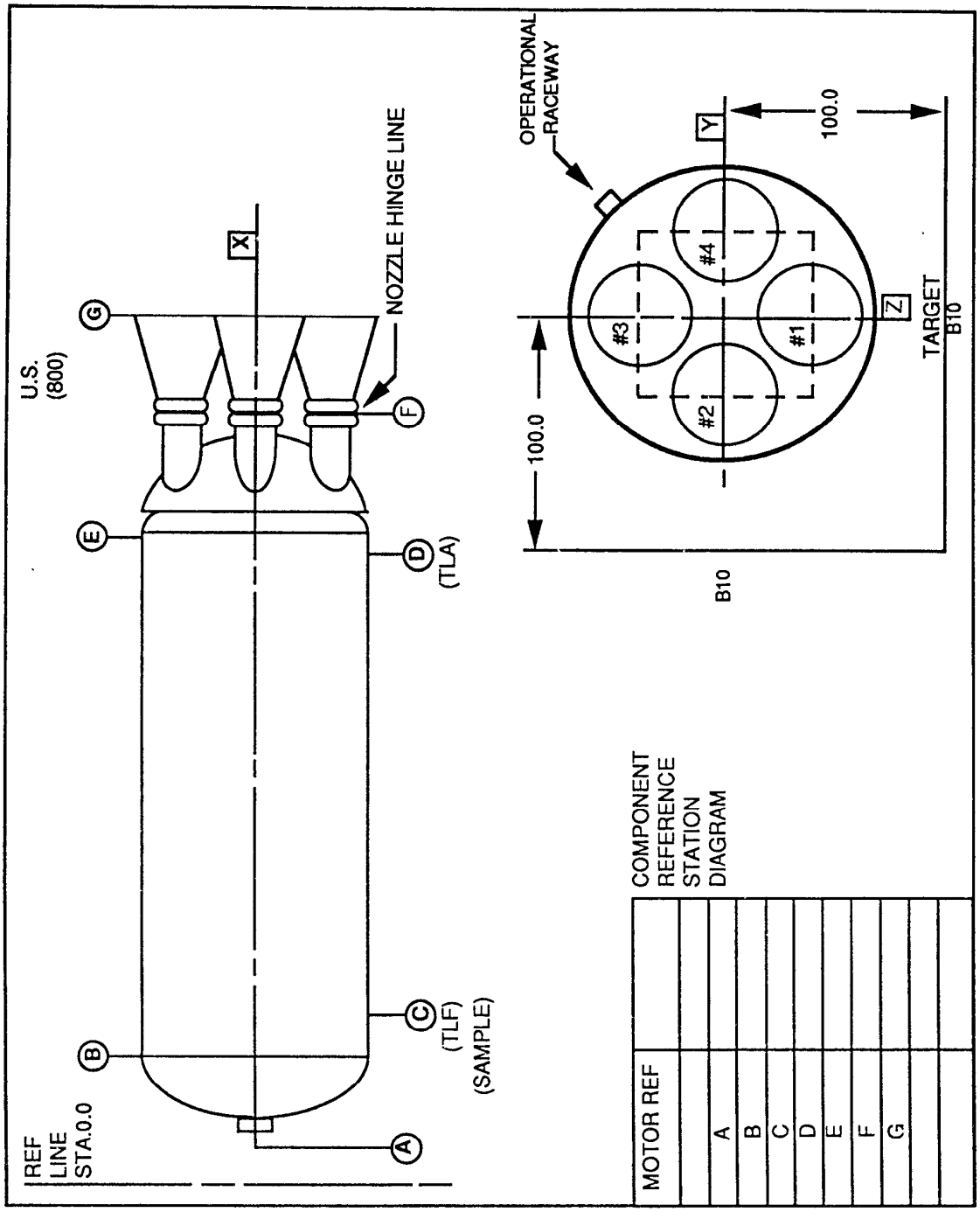


FIGURE 2. Sample component reference station diagram and center of gravity dimensional description

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Block 10. Preparation Instructions (Continued)

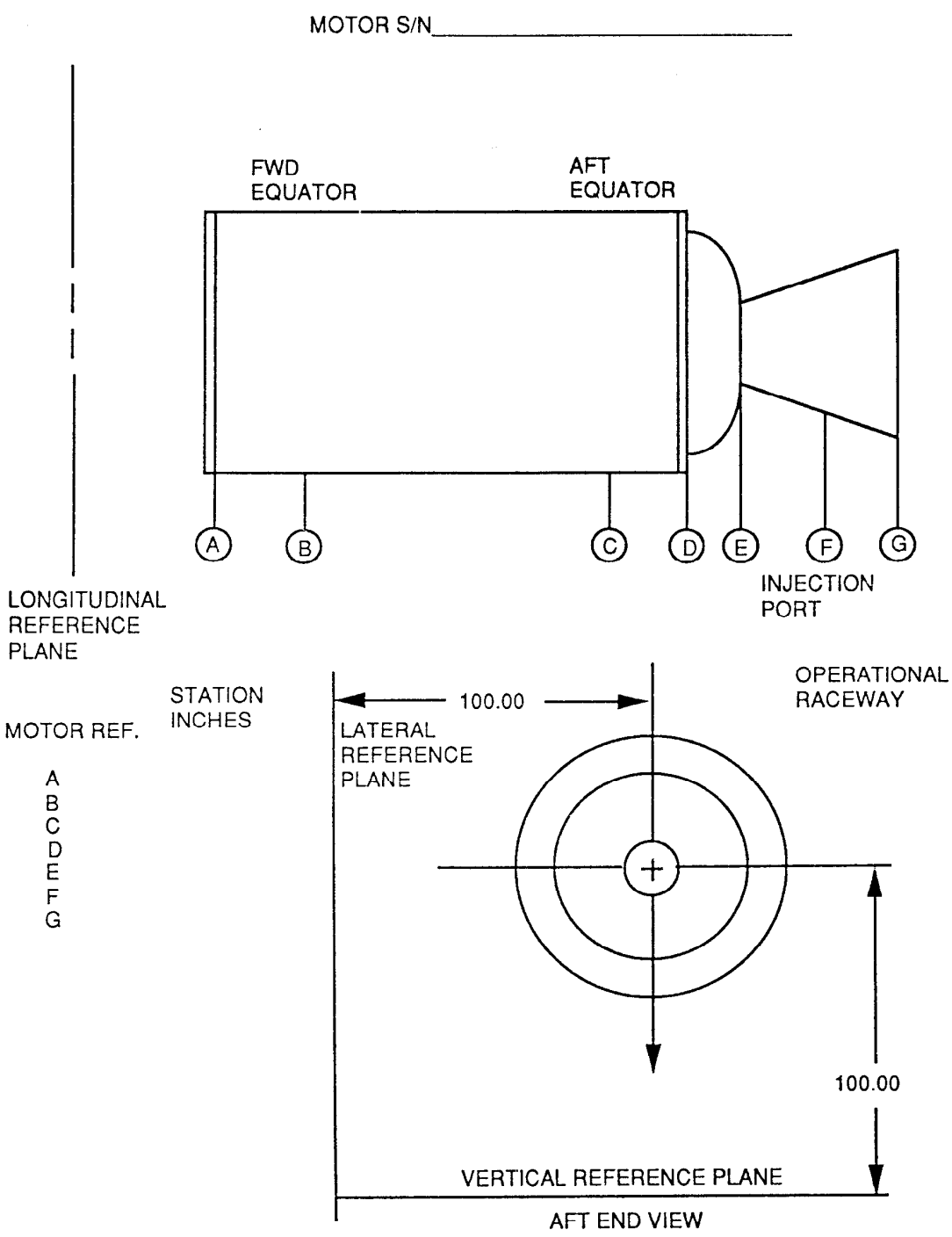


FIGURE 2. Sample component reference station diagram and center of gravity dimensional description - Continued

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Block 10. Preparation Instructions (Continued)

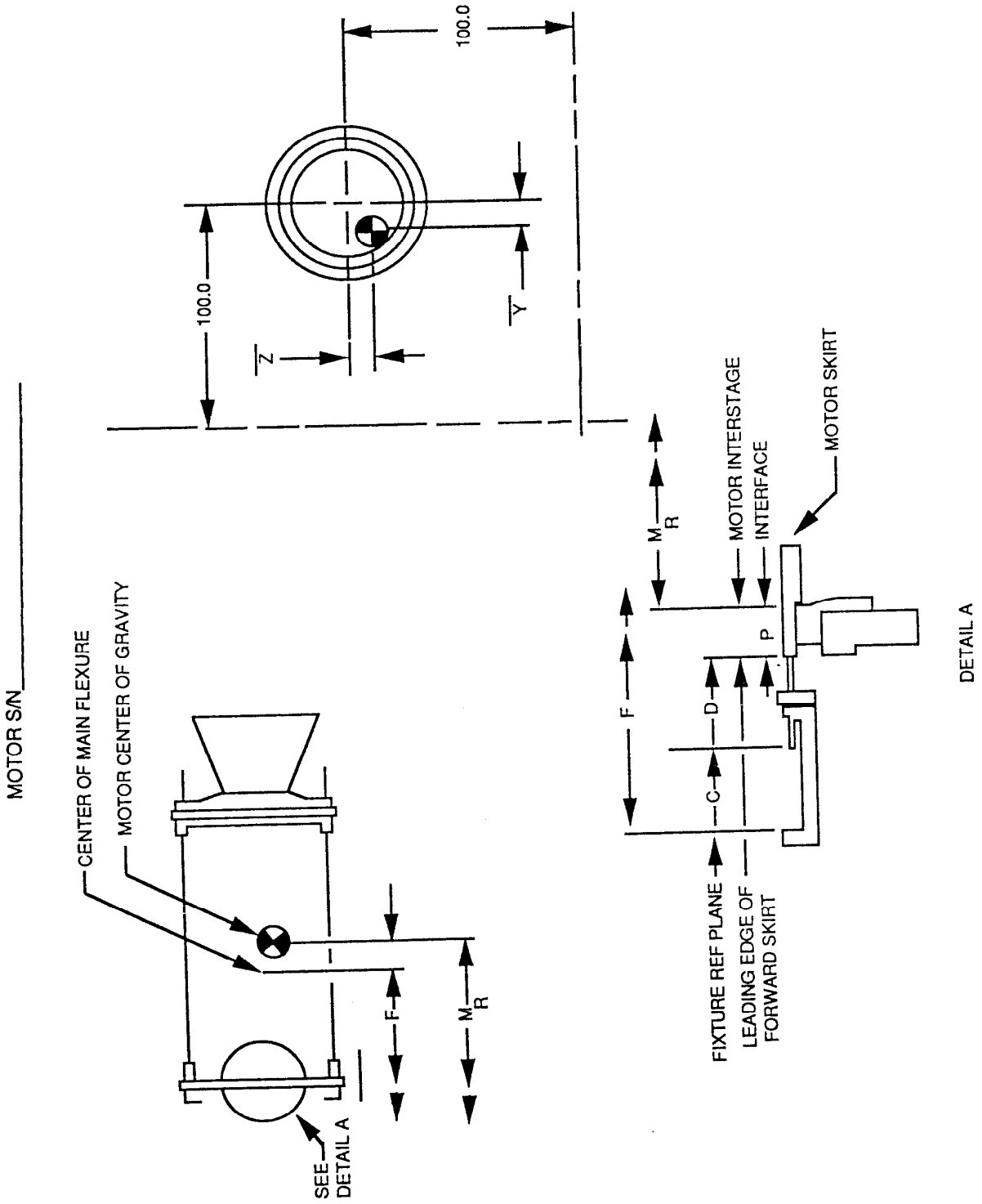


FIGURE 2. Sample component reference station diagram and center of gravity dimensional description - Continued

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Block 10. Preparation Instructions (Continued)

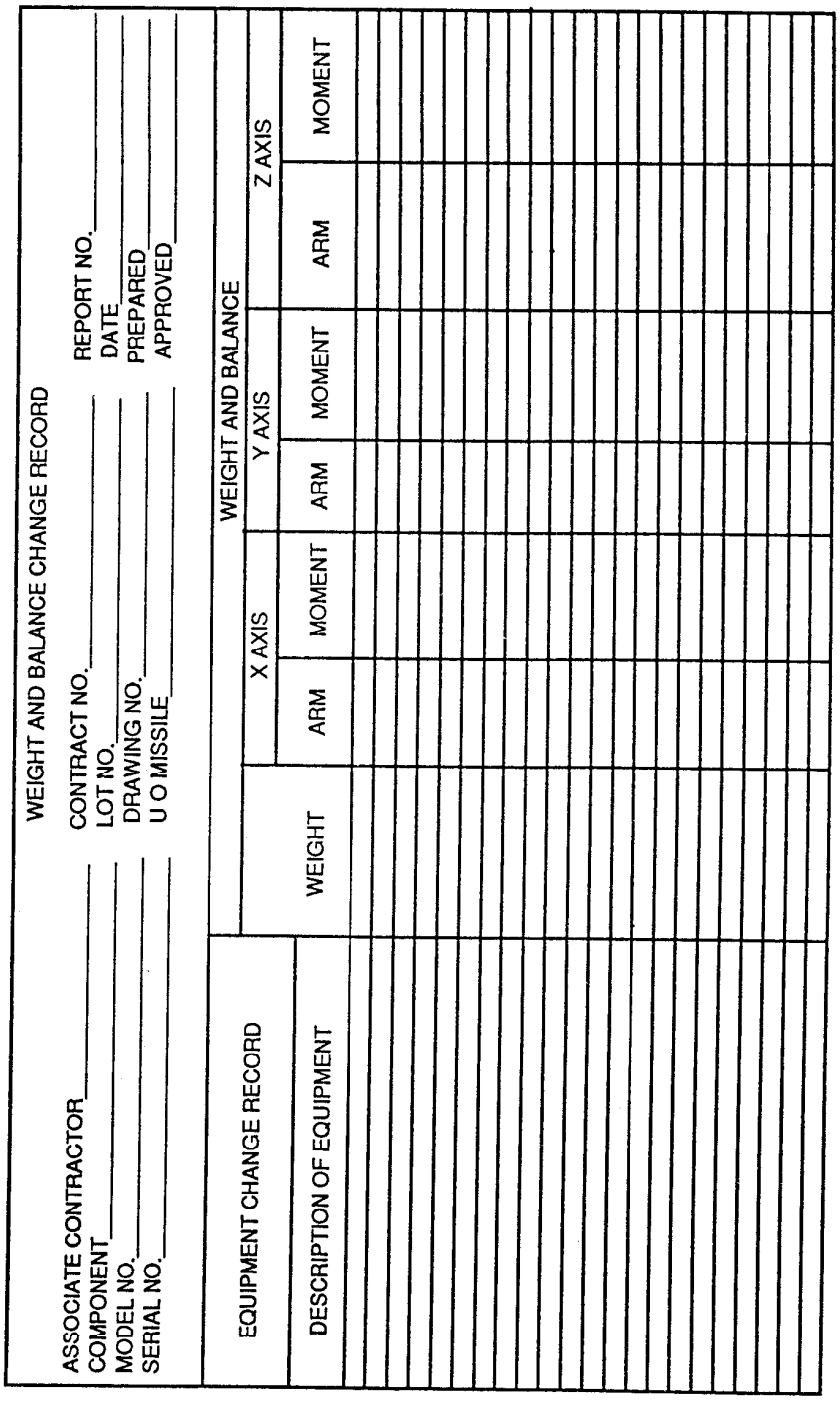


FIGURE 3. Sample weight and balance change record

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Block 10. Preparation Instructions (Continued)

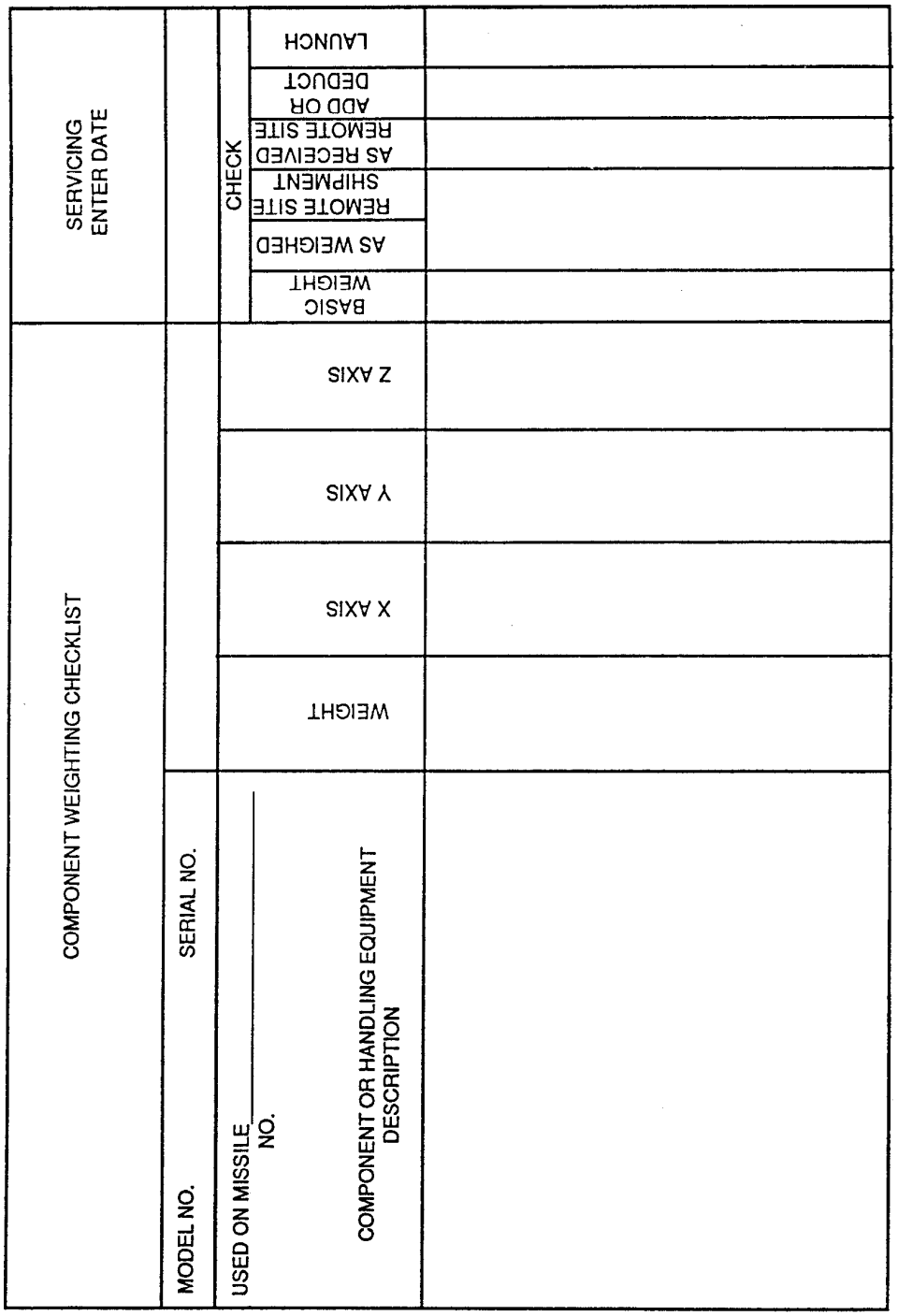


FIGURE 4. Sample component weighting checklist

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Block 10. Preparation Instructions (Continued)

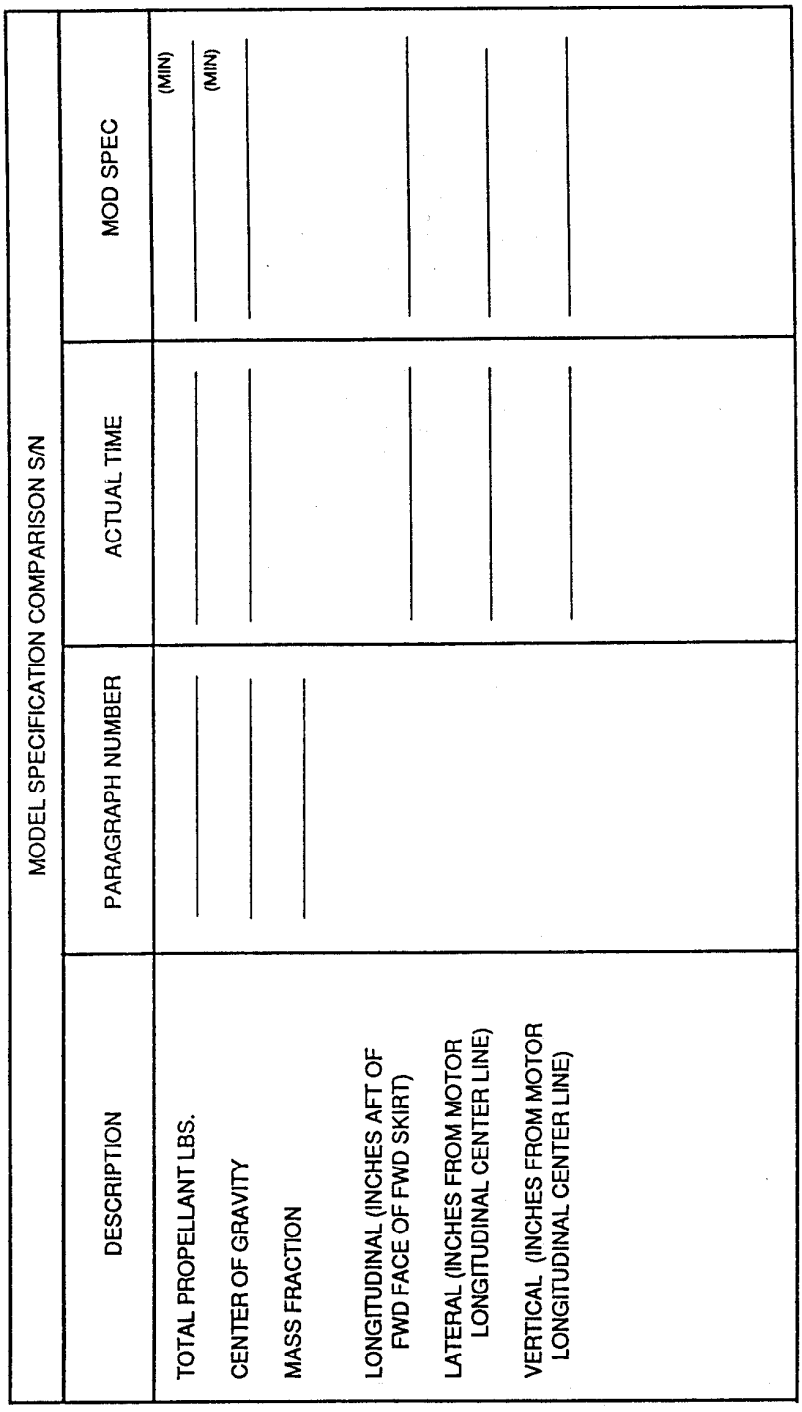


FIGURE 5. Sample model specification comparison S/N

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Block 10. Preparation Instructions (Continued)

TRANSFORMER CALIBRATION REPORT
OPERATIONAL PRESSURE

MANUFACTURE	SERIAL NO. DATE
MODEL NO.	RANGE PSIA
EXCITER VOLTAGE	TEMPERATURE
BAROMETER READING	

APPLIED LOAD		OUTPUT SIGNAL		OUTPUT IMPEDANCE	OHMS
% OF FULL SCALE	PSIG	% OF FULL SCALE		ISOLATION PINS 2L	MEGHOMS
		ASCENDING	DECENDING	RESISTANCE PINS TO CASE	MEGHOMS
0	0			ELECTRICAL CALIBRATION SIGNAL IS:	% FS
10	100			OUTPUT AT ZERO PSIG:	VOLTS
25	250			OUTPUT AT 1000 PSIG	VOLTS
50	500			SPAN	VOLTS
75	750			MAXIMUM NON-LINEARITY	% FS
90	900			MAXIMUM HYSTERESIS	% FS
100	1000			MAXIMUM NON-REPEATABILITY	% FS

APPROVED

MOTOR NO.	MEASUREMENT
MISSILE NO.	MECHANICAL INSTALLATION NO.

SIGNATURE

FIGURE 6. Sample transducer calibration report

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MODEL_____REV_____DATE_____

DWG NO._____TOTAL NO CYCLES_____

SERIAL NO._____PREVIOUS NO TEST CYCLES_____

LOT I.D._____NO CYCLES THIS INSP_____

DATE OF INSPECTION_____CYCLE TEMPERATURE_____

DATE OF MANUFACTURE_____

"SAFE POSITION"	SPEC PARA REF	REQUIRED	18 V 1	30 V 2	18 V 3	30 V 4	TECH	INSP
ARM COMMAND RESISTANCE								
SAFE COMMAND RESISTANCE								
ARM MONITOR CIR RESISTANCE								
SAFE MONITOR CIR RESISTANCE								
SQUIB SIM 1 RESISTANCE								
SQUIB SIM 2 RESISTANCE								
ALL CIR TO CASE RESISTANCE								
COM CIR TO FIR CIR								
ARMING TIME (MAX)								
N/A								
N/A								
"ARM POSITION"								
ARM COMMAND CIR RESISTANCE								
SAFE COMMAND CIR RESISTANCE								
ARM MONITOR CIR RESISTANCE								
SAFE MONITOR CIR RESISTANCE								
SQUIB SIM 1 RESISTANCE								
SQUIB SIM 2 RESISTANCE								
ALL CIRCUITS TO CASE RESISTANCE								
CONT CIR TO FIR CIR								
ARMING TIME (MAX)								
N/A								
N/A								

MINIMUM ARMING VOLTAGE_____

MINIMUM SAFING VOLTAGE_____

WEIGHT WITH COVER_____

LEAK RATE_____

Figure 7. Sample standardized igniter safe arm mechanism inspection report

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Block 10. Preparation Instructions (Continued)

MODEL_____REV_____DATE_____

DWG NO._____TOTAL NO CYCLES_____

SERIAL NO._____PREVIOUS NO TEST CYCLES_____

LOT I.D._____NO CYCLES THIS INSP_____

DATE OF INSPECTION_____CYCLE TEMPERATURE_____

DATE OF MANUFACTURE_____

"SAFE POSITION"	SPEC PARA REF	REQUIRED	∇ 1	∇ 2	∇ 3	∇ 4	TECH	INSP
1. ARM COMMAND CIRCUIT RESISTANCE								
2. SAFE MOTOR COIL LEAK CURRENT								
3. ARM MONITOR CIRCUIT								
4. SAFE MONITOR CIRCUIT RESISTANCE								
5. SQUIB SIM 1 CIRCUIT								
6. SQUIB SIM 2 CIRCUIT								
7. SQUIB SIM 3 CIRCUIT								
8. SQUIB SIM 4 CIRCUIT								
9. ALL EXTS TO CASE INSULATION								
10. CONTROL CIRCUIT TO FIRE CIRCUIT								
11. ARMING TIME (MAX)								
"ARM POSITION"								
12. ARM MOTOR COIL LEAK CURRENT								
13. SAFE MOTOR COIL RESISTANCE								
14. ARM MONITOR CIRCUIT RESISTANCE								
15. SAFE MONITOR CIRCUIT RESISTANCE								
16. SQUIB 1 RESISTANCE								
17. SQUIB 2 RESISTANCE								
18. SQUIB 3 RESISTANCE								
19. SQUIB 4 RESISTANCE								
20. ALL CIRCUITS TO CASE INSULATION								
21. CONTROL CIRCUIT TO FIRE CIRCUIT								
22. SAFING TIME								

Figure 8. Sample standardized thrust termination switch inspection report

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